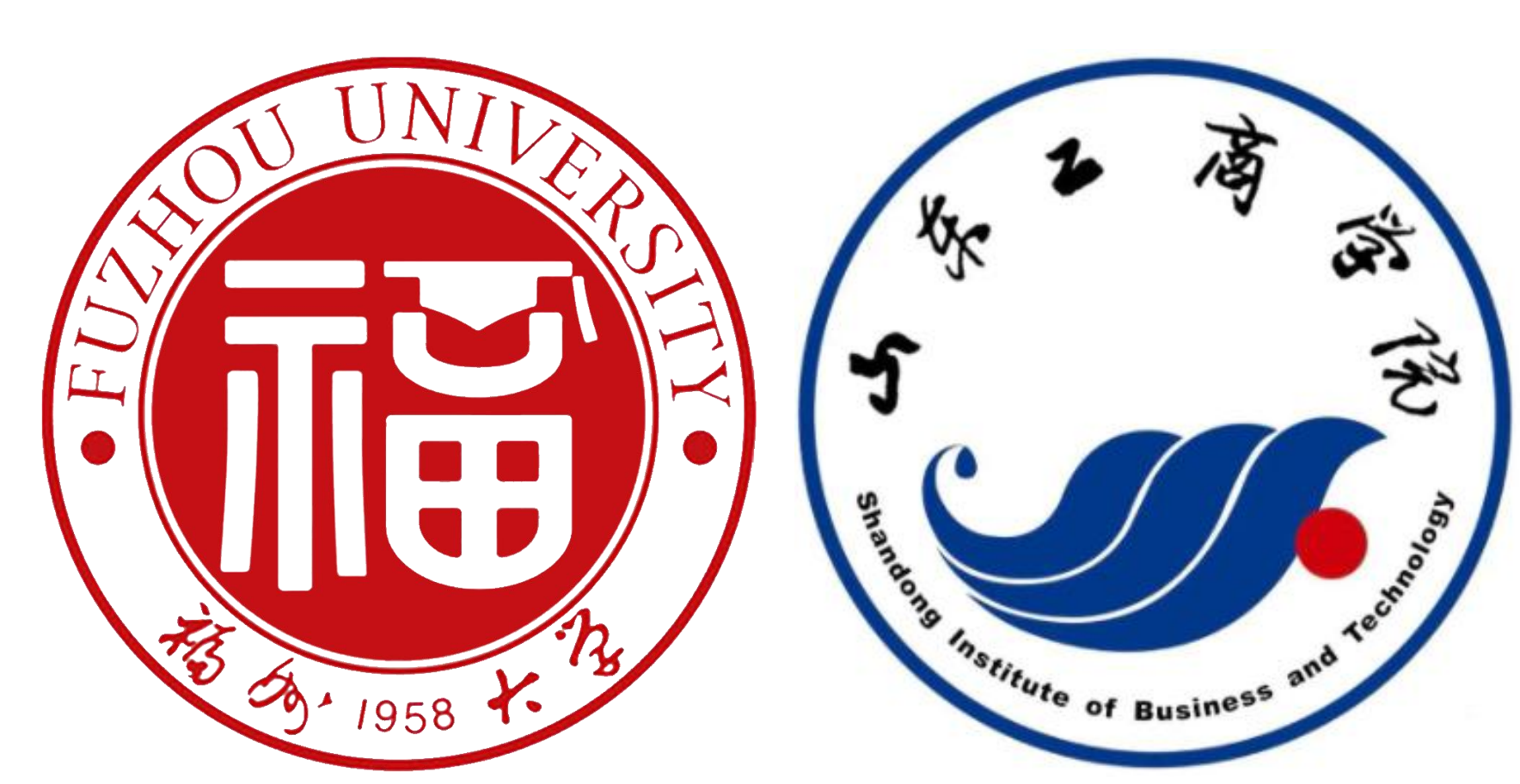


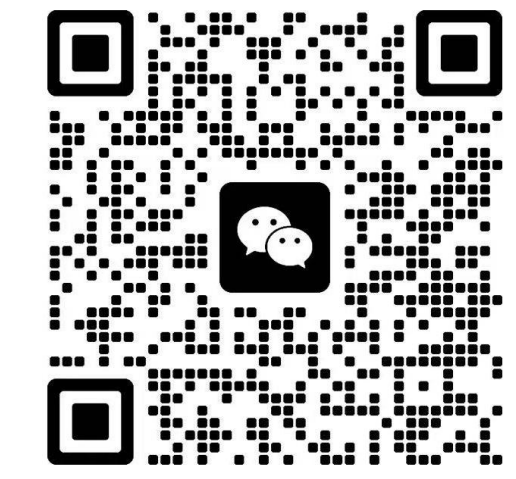


BiEvLight: Bi-level Learning of Task-Aware Event Refinement for Low-Light Image Enhancement

Zishu Yao¹, Xiang-Xiang Su¹, Shengning Zhou², Guang-Yong Chen¹, Guodong Fan^{2*}, Xing Chen¹

¹ Fuzhou University, China

² Shandong Technology and Business University, China



Contact us



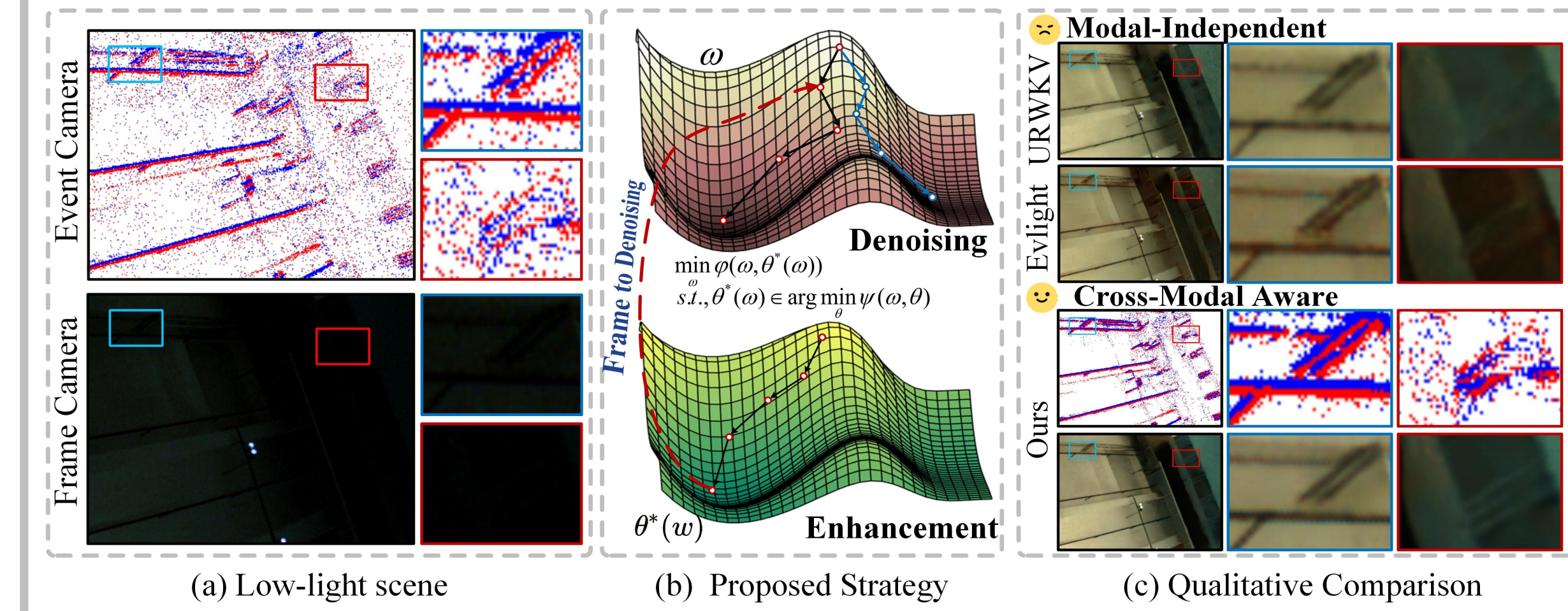
Project

CVPR
JUNE 3-7, 2026



DENVER
COLORADO

Motivation



Problem:

Dual degradation in **low-light** scenes: **Noisy** frames + **BA**-corrupted events.

Limitation of existing methods:

* **No image-guided denoising**: BA noise is hard to separate from structural events.

* **No task-aware feedback**: Refined events may not benefit LLIE.

Insight:

Task-aware event refinement and image enhancement form a **collaborative optimization** process.

Contributions

- **Task-aware bilevel framework** for joint event denoising and LLIE.
- **Gradient-guided event refinement** using **image-event structural correlation**.
- **State-of-the-art** results on real-world low-light event-image benchmarks.

Methods

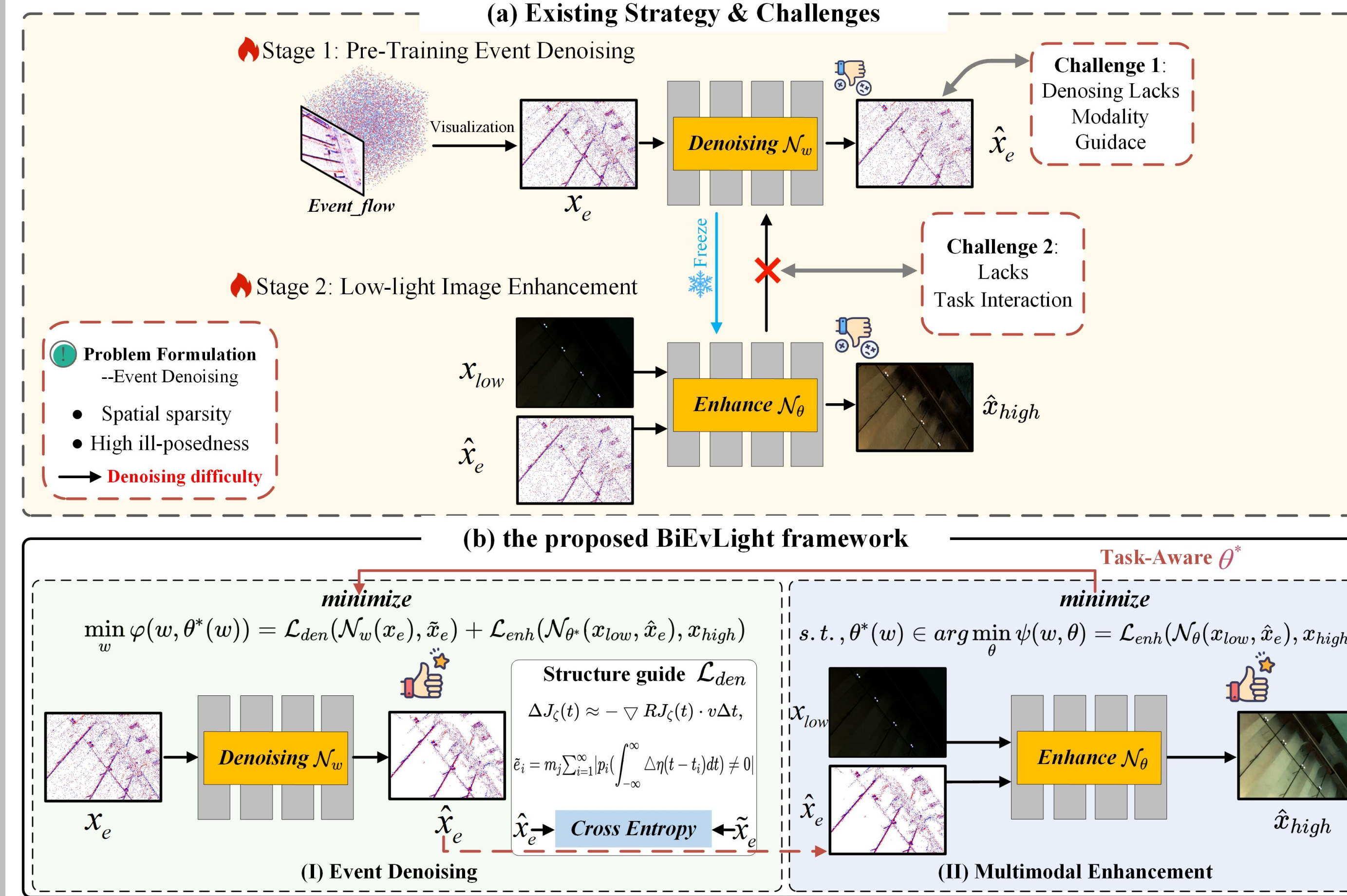


Fig. 2. (a) Problems and challenges of existing methods, (b) The proposed BiEvLight framework.

Optimization Algorithm

Algorithm 1 Bi-level Learning of Task-Aware

Input: The initial parameter w_0, θ_0 , step-sizes η_θ, η_w , gradient supervision level μ , iteration numbers K .

Output: The optimal parameters w^*, θ^* .

```

1: for  $k = 0 : K - 1$  do
2:   % Update the lower-level parameter.
3:    $\theta_{k+1} = \theta_k - \eta_\theta \nabla_\theta \psi(w_k, \theta_k)$ .
4:   Calculate the upper-level gradient by (20)-(21).
5:   % Update the upper-level parameter.
6:    $w_{k+1} = w_k - \eta_w \nabla_w \phi(w_k, \theta_{k+1})$ .
7: end for

```

Task-Aware Refinement

$$\hat{x}_e = \mathcal{N}_w(x_e), \quad \hat{x}_{high} = \mathcal{N}_\theta(x_{low}, \hat{x}_e)$$

Enhancement-Constrained Update

$$\theta_{k+1} = \theta_k - \eta_\theta \nabla_\theta \psi(w_k, \theta_k)$$

$$w_{k+1} = w_k - \eta_w \nabla_w \phi(w_k, \theta_{k+1})$$

Results

Quantitative comparison

Datasets		Image Only						Image + Event			
		SNRNet	FourLLIE	Reformer	NeRCO	SCI++	URWKV	eSLNet	ELIE	EvLight	BiEvLight
Reference		CVPR' 22	ACM MM' 23	ICCV' 23	ICCV' 23	TPAMI' 25	CVPR' 25	ECCV' 20	TMM' 23	CVPR' 24	
SDE-in	PSNR $\uparrow$	20.3482	19.7856	21.3169	19.4266	12.4683	21.5831	21.3622	22.0414	22.1880	22.8680
	PSNR* $\uparrow$	23.6785	23.2709	23.6502	23.2052	22.3026	22.6191	23.0825	23.6395	23.6940	26.0023
	SSIM $\uparrow$	0.6267	0.6176	0.6873	0.5626	0.4928	0.7168	0.7023	0.7083	0.7189	0.7750
SDE-out	PSNR $\uparrow$	21.5542	20.6323	22.3075	18.4125	12.0277	22.3732	20.6244	22.2167	22.4372	24.3599
	PSNR* $\uparrow$	23.3224	22.1990	23.9459	22.2225	23.2990	24.0964	22.3612	23.6772	24.4223	26.1617
	SSIM $\uparrow$	0.6508	0.6410	0.6981	0.5490	0.3033	0.6880	0.6295	0.7002	0.7070	0.7451
SDSD-in	PSNR $\uparrow$	25.9892	24.2775	27.7175	19.9693	9.9750	28.7034	25.8946	28.3385	29.3563	30.7576
	PSNR* $\uparrow$	26.4070	28.0437	28.2613	23.4650	29.3317	28.9094	26.3361	29.6009	30.4038	30.9992
	SSIM $\uparrow$	0.8730	0.8668	0.9125	0.7821	0.3108	0.9158	0.8961	0.9187	0.9250	0.9473
SDSD-out	PSNR $\uparrow$	23.6656	23.4002	26.3934	21.7971	18.9373	22.0251	23.3921	25.0251	26.7407	27.4108
	PSNR* $\uparrow$	27.0967	27.7418	28.6221	27.6637	28.8727	27.2018	26.3331	29.1202	30.3066	30.4579
	SSIM $\uparrow$	0.8310	0.7944	0.8485	0.7482	0.5776	0.8176	0.8031	0.8626	0.8673	0.8873

Tab. 1. Quantitative comparison with advanced methods of low-light image enhancement on SDE and SDSO dataset. The red and blue represent the best and second-best values.

Qualitative comparison

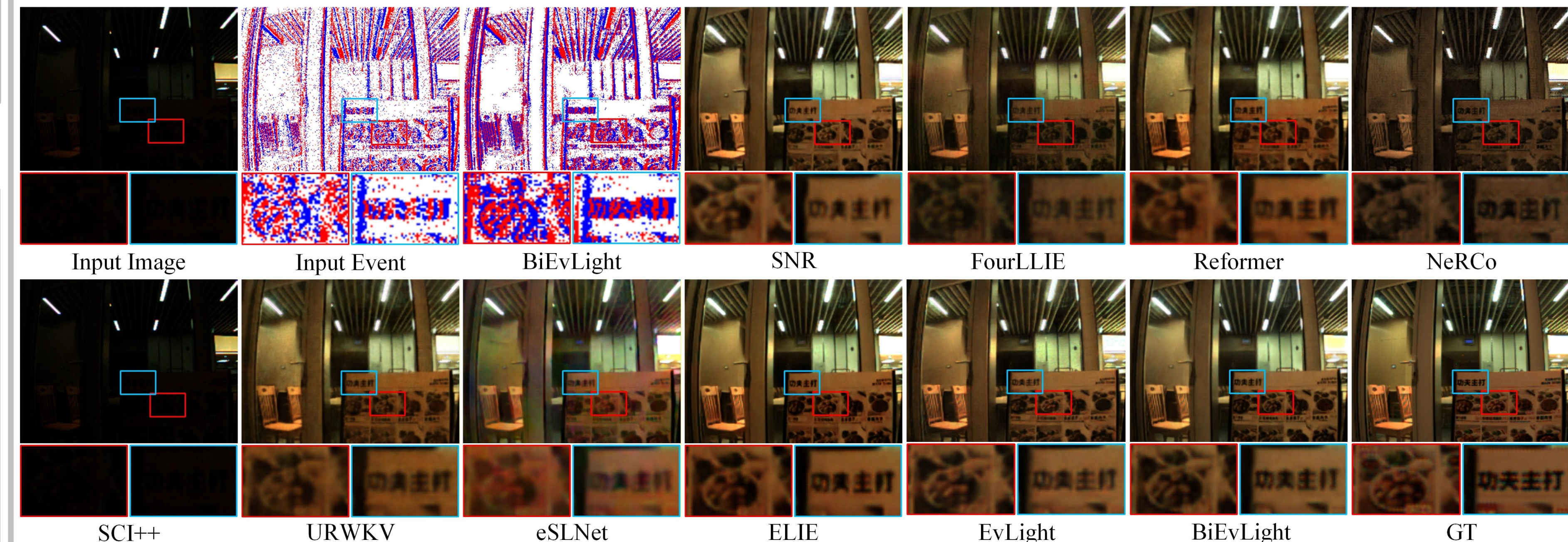


Fig. 3. Quantitative comparison with advanced methods of low-light image enhancement on SDE and SDSO dataset. The red and blue represent the best and second-best values.